

Mic Drop

Final Project Spec

A local karaoke party game played over a single phone call. Sign up with your Solana wallet, create a game, invite friends, pass one phone around, and sing. Scores are deterministic, the leaderboard is live, and the pot pays out on-chain.

Tracks: ElevenLabs · Solana · Snowflake · Google/Gemini · Twilio **Time:** 36–48 h **Team:** 3–4, comfortable with Solana

1 The one principle everything hangs on

The system is split into two layers that never cross:

Layer	Owns	Components	Property
Money / fairness	the score and the pot	deterministic scorer + Solana	reproducible, auditable, undisputable
Fun / personality	the show	Gemini (director) + ElevenLabs (voice)	creative, varied, can regenerate freely

Gemini orchestrates and narrates; the scorer owns the number; Solana owns the pot. You can regenerate the host’s banter a hundred times and the score never moves. That split is why it is safe to put a pot on the line — and it is the cleanest thing to tell judges.

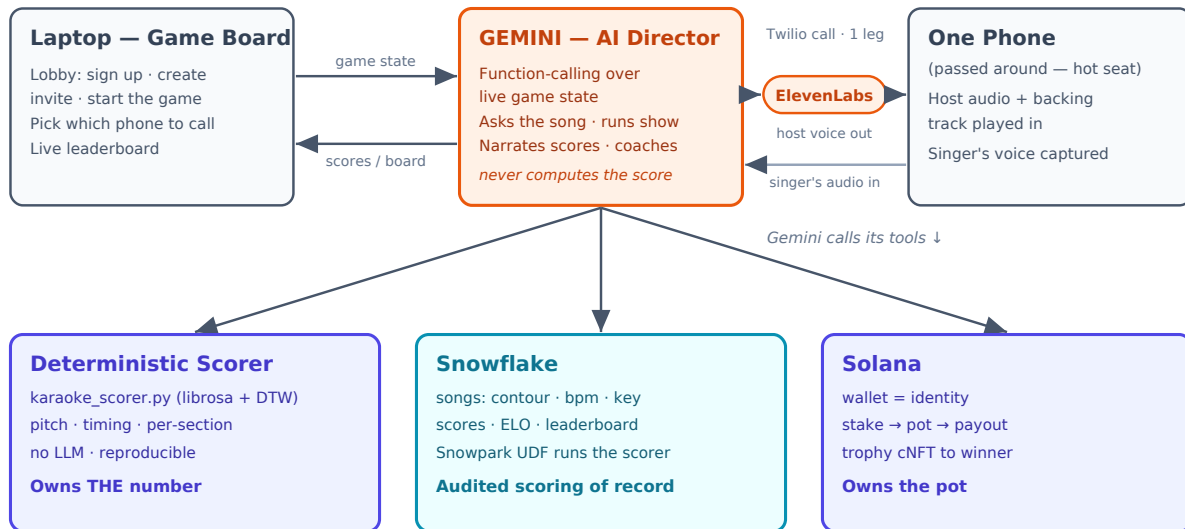
2 Locked decisions

- **Karaoke only** — one game type, done well.
- **Hot-seat, one shared phone (Option A)** — local play. Pick one phone at game start, dial it once, players pass it around for their turns.
- **Single continuous call, segmented by turn** — one voice at a time = one clean clip per turn. No conference, no stream separation, no mute juggling.
- **Laptop = the game board** (lobby + live leaderboard); **phone = the mic + host channel**.
- **Scoring is deterministic and already built** (`karaoke_scorer.py`).
- **Gemini = AI game director**, not a scorer.

3 Architecture

Mic Drop — System Architecture

Karaoke over one phone call · five sponsor technologies · two strictly separated layers



- **Fun / personality** — creative, can vary freely (Gemini director + ElevenLabs voice)
- **Money / fairness** — deterministic & auditable (Scorer owns the number, Solana owns the pot)
- **Data / infra** — Snowflake (songs, scores, audit) and Twilio (the single call)

4 App flow

4.1 Lobby (on the laptop)

1. **Sign up** → username + SOL address + phone (stored per user).
2. **Create game** → a room the creator runs on the laptop.
3. **Invite players** → add registered users to the roster (who plays / whose wallet stakes / who is on the leaderboard). Optional: SMS the invite via Twilio.
4. **Start game** → stakes lock from each player's wallet into the pot.
5. **"Whose phone do we call?"** → pick one number; Twilio dials it; the AI host picks up.

4.2 Per-turn loop (on the call)

1. Gemini (as host) **asks the player what song** they want → validates against the song library.
2. Gemini calls `start_turn` → backing track + host intro play into the call.
3. Backend captures the singer's audio for the song duration (one clean stream).
4. `extract_pitch_midi` → `score_karaoke` vs the song's reference contour → score JSON (incl. per-section breakdown).
5. Gemini `reveal_score` (narrates the number) + `give_coaching` (specific feedback). Written to Snowflake; laptop leaderboard updates.

6. Pass the phone → next player → repeat.
7. After all turns / best-of-N: Gemini `declare_winner` → Solana pays the pot + mints a **trophy cNFT** → end-of-game recap.

5 Component specs

5.1 Gemini — the AI game director

An agent with tools it calls over live game state. It **decides and announces; the systems execute**. It never computes the score.

Tool	Gemini's job	Executes
<code>ask_song(player)</code>	prompt the player by voice, capture choice	validate vs library
<code>pick_song(...)</code>	<i>suggest</i> a song from history / range / keep-it-close	reads Snowflake
<code>set_host_mode(...)</code>	choose persona (roast / encourager / hype)	tone only
<code>apply_twist(...)</code>	pick + announce a round modifier from a fixed menu	maps to a deterministic scorer parameter
<code>start_turn(...)</code>	kick off the turn	triggers Twilio backing track + capture
<code>reveal_score(...)</code>	narrate the deterministic number	reads scorer JSON (read-only)
<code>give_coaching(...)</code>	actionable feedback	reads numbers + metadata
<code>update_banter(...)</code>	between-song patter, react to standings	language only
<code>declare_winner(...)</code>	call the match, narrate the recap	triggers Solana payout + cNFT

Gemini works purely from **numbers + text** (score JSON + song metadata + game state). It never touches audio — keeping it fast and keeping the deterministic story clean. Core build: director + MC patter + coaching + personas. Stretch: smart song selection, end-of-game recap, crowd-work.

5.2 Deterministic scorer — `karaoke_scorer.py` **[built]**

No LLM, no randomness — `librosa.pyin` + DTW + `numpy`, reproducible with pinned params (`SCORER_VERSION`). The pitch contour is the compact feature (“the bits”) all scoring runs on, so any score can be recomputed from stored features.

- **Inputs:** player pitch contour, reference (contour + sections + beat_grid), optional transcript + reference lyrics, optional deterministic twist modifier.
- **Output JSON:**

```
{ "total": 72.0, "pitch": 85.0, "on_pitch_pct": 64.0, "timing": 41.0,
  "sections": [ { "name": "verse 1", "pitch": 90, "timing": 70 },
                 { "name": "chorus", "pitch": 80, "timing": 25 } ],
  "lyrics": null, "octave_shift": 0.0, "scorer_version": "karaoke-1.0.0" }
```

- **Properties:** octave-equivalent (low and high voices both score fairly); pitch + timing are the MVP; lyrics optional (needs upstream STT).
- **Planned extension:** per-section sub-scores + beat-grid-based timing (replaces the DTW-warp proxy). Those per-section numbers feed Gemini's specific coaching.

5.3 ElevenLabs — the host’s voice

Renders whatever Gemini writes into the game-show host’s voice, played into the Twilio call. Words come from Gemini; voice comes from ElevenLabs; they are a pipeline, not an overlap.

5.4 Snowflake — data + audited scoring

- `songs` table (schema below) — reference contours + musical metadata.
- Scores / rounds / ELO + the live leaderboard.
- `score_karaoke` runs as a **Snowpark UDF** so every score is auditable and recomputable from the stored contour — the scoring engine of record.

5.5 Solana — identity, stakes, trophies

Wallet = identity (collected at signup). Stake → pot at game start → winner payout. Champion mints a **trophy cNFT** (cheap, compressed). *Demo*: custodial pot on devnet; name the production path (on-chain escrow program).

5.6 Twilio — the single call

One outbound call to the chosen phone: host audio + backing-track playback + per-turn capture. SMS invites. **Gotcha: trial accounts only dial verified numbers** — verify all demo phones on day one.

6 Song metadata (Snowflake `songs` table)

One record feeds both layers; different fields go to different places.

Field	Goes to	Purpose
<code>song_id</code> , <code>title</code> , <code>artist</code>	both / Gemini	identity, banter, recap
<code>reference_contour</code>	scorer	ground-truth pitch over time
<code>bpm</code> , <code>beat_grid</code>	scorer	real beat-based timing scoring
<code>sections</code>	scorer	per-section sub-scores
<code>backing_track_ref</code>	Twilio	audio played into the call
<code>key</code>	Gemini	“you flatted relative to G major”
<code>vocal_range</code> , <code>hardest_passage</code>	Gemini	pinpoint difficulty in feedback
<code>genre</code> , <code>mood</code> , <code>difficulty</code>	Gemini	persona, banter, song selection
<code>reference_lyrics</code>	scorer (opt.)	lyric scoring if STT added

Automation: during reference prep, auto-extract `bpm` (`librosa.beat.beat_track`), `vocal_range` (contour min/max), and approximate `key` (chroma) from the “correct take” — so adding a song is mostly automatic.

7 Build plan (36–48 h)

Phase 0 — Setup (2–3 h): accounts (Twilio + *verify demo numbers*, ElevenLabs, Snowflake trial, Gemini, Solana devnet); Snowflake tables; devnet wallets + pot stub. (*scorer already done*)

Phase 1 — The call loop (~10 h, long pole): one Twilio outbound call → play host audio + backing track

→ capture a singer's turn → save clip. One full turn end-to-end.

Phase 2 — Scoring + songs (~8 h): wire clip → extract_pitch_midi → score_karaoke → Snowflake. Extend scorer to per-section + beat timing. Reference-prep script + 2–3 songs.

Phase 3 — Gemini director + lobby (~10 h): function-calling director (ask song, banter, reveal, coaching, personas) + ElevenLabs voice; laptop lobby (signup / create / invite / start / pick-phone) + live leaderboard.

Phase 4 — Solana + polish (~8 h): stake / pot / payout + trophy cNFT; end recap; error handling; **record a backup full run**; rehearse 3×.

Buffer + sleep: protect ~4–6 h.

7.1 Roles

- **P1 Telephony** — Twilio call, backing-track playback, per-turn capture, turn control.
- **P2 AI** — Gemini director + tool schema, ElevenLabs voice, coaching / persona prompts.
- **P3 Chain** — Solana wallet auth, stake / pot / payout, trophy cNFT.
- **P4 Data + Frontend** — Snowflake + Snowpark UDF, songs / reference prep, lobby web app, live leaderboard, demo.

8 Demo script (~3 min)

1. **(15s)** “A karaoke party game over one phone call — real scores, real stakes on Solana. Let’s play.”
2. **(20s)** Two presenters sign up (username + SOL + phone), create a game, stake devnet USDC. Start → “call this phone” → it rings, the AI host picks up.
3. **(70s)** Host asks Player 1 their song → backing track plays → they sing → host banter → **score + coaching land on the leaderboard**. Pass the phone → Player 2 → score lands.
4. **(25s)** Host declares the winner → **trophy cNFT hits their wallet** (show explorer) → pot pays out. Snowflake leaderboard on screen.
5. **(10s)** “Twilio, ElevenLabs, Gemini, Solana, Snowflake — one phone, one game.”

Backup: pre-recorded full run, queued (live calls misbehave at the worst moment).

9 Gotchas (handle early)

- **Twilio trial = verified numbers only** — verify every demo phone day one or “start game” silently fails.
- **Backing-track timing** — reference contour must align to the same backing track players hear, or timing scoring drifts.
- **Phone audio is 8 kHz** — pitch scoring still works; do not expect studio fidelity.
- **Lyrics scoring is optional** — MVP is pitch + timing (already a complete karaoke score).
- **Pot security** — custodial devnet for the demo; name the production escrow path.
- **Keep Gemini away from the score** — even twists route through deterministic parameters.

10 Status & scope

Done: deterministic scorer (pitch / timing / optional lyrics, octave-equivalent, reproducible).

To build: scorer per-section + beat extension · Twilio call loop · reference prep + songs table · Gemini director + ElevenLabs voice · lobby + leaderboard · Solana pot / trophy.

Out of scope (future): remote two-phone play · “Clone Wars” voice-clone round · full STT lyrics scoring.